

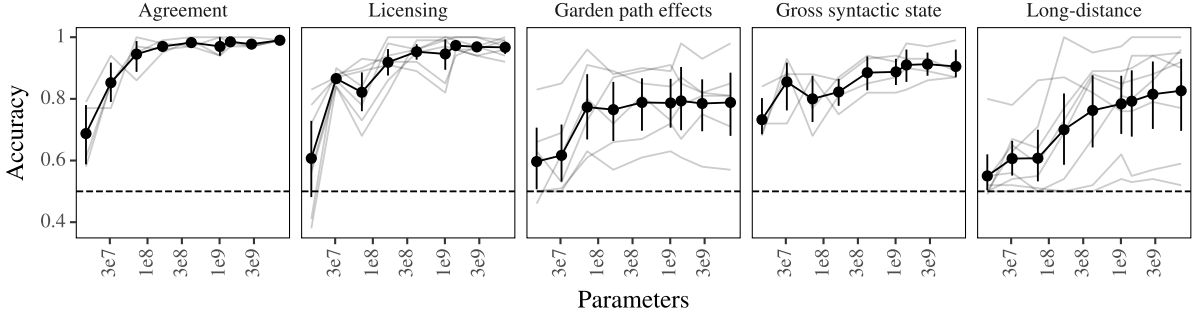


Figure 3: Accuracy of pythia-family models on the **CausalGym** tasks, grouped by type, with scale. The dashed line is random-chance accuracy (50%).

3.3 Tasks

CausalGym contains 29 tasks, of which one is novel (`agr_gender`) and 28 were templatised from **SyntaxGym**. Of the 33 test suites in the original release of **SyntaxGym**, we only used tasks from which we could generate paired grammatical sentences (leading us to discard the 2 center embedding tasks), and merged the 6 gendered reflexive licensing tasks into 3 non-gendered ones. We show task accuracy vs. model scale in Figure 3. Examples of pairs generated for each task are provided in appendix A.

3.4 Evaluation

An evaluation sample consists of a base input $\mathbf{b}$, source input $\mathbf{s}$, ground-truth base label y_b , and ground-truth source label y_s . For example, the components of (6) are

- (7) $\underbrace{\text{The author near the senators}}_{\mathbf{b}} \Rightarrow \underbrace{\text{is}}_{y_b}$
 (8) $\underbrace{\text{The authors near the senators}}_{\mathbf{s}} \Rightarrow \underbrace{\text{are}}_{y_s}$

A successful intervention will take the original LM p running on input $\mathbf{b}$ and make it predict y_s as the next token. We measure the strength of an intervention by its **log odds-ratio**.

First, we select a component f , which can be any part of a neural network that outputs a representation, inside the model p . When the model is run on input $\mathbf{b}$, this component produces a representation we denote $f(\mathbf{b})$. We perform an intervention which replaces the output of f with an output of f^* as in §2. To produce a representation, f^* may modify the base representation with reference to the source representation, and so its output is $f^*(\mathbf{b}, \mathbf{s})$. The intervention results in an intervened language model which we denote informally as $p_{f \leftarrow f^*}$. In the framework of causal abstraction (Geiger et al.,

2021), if this intervention successfully makes the model behave as if its input was $\mathbf{s}$, then the representation at f is causally aligned with the high-level linguistic feature alternating in $\mathbf{b}$ and $\mathbf{s}$.

We now operationalise a measure of causal effect. Taking the original model p , the intervened model $p_{f \leftarrow f^*}$, and the evaluation sample, we define the log odds-ratio as:

$$\text{Odds}(p, p_{f \leftarrow f^*}, \langle \mathbf{b}, \mathbf{s}, y_b, y_s \rangle) = \log \left(\frac{p(y_b | \mathbf{b})}{p(y_s | \mathbf{b})} \cdot \frac{p_{f \leftarrow f^*}(y_s | \mathbf{b}, \mathbf{s})}{p_{f \leftarrow f^*}(y_b | \mathbf{b}, \mathbf{s})} \right) \quad (9)$$

where a greater log odds-ratio indicates a larger causal effect at that intervention site, and a log odds-ratio of 0 indicates no causal effect. Given an evaluation set E , the average log odds-ratio is

$$\text{AvgOdds}(p, p_{f \leftarrow f^*}, E) = \frac{1}{|E|} \sum_{e \in E} \text{Odds}(p, p_{f \leftarrow f^*}, e) \quad (10)$$

4 Methods

We briefly describe our choice of f^* and the feature-finding methods that we benchmark in this paper.

4.1 Preliminaries

In this paper, we only benchmark interventions along a single feature direction, i.e. one-dimensional distributed interchange intervention (1D DII; Geiger et al., 2023b). DII is an interchange intervention that operates on a non-basis-aligned subspace of the activation space. Formally, given a feature vector $\mathbf{a} \in \mathbb{R}^n$ and f , 1D DII defines f^* as

$$f^*(\mathbf{b}, \mathbf{s}) = f(\mathbf{b}) + (f(\mathbf{s})\mathbf{a}^\top - f(\mathbf{b})\mathbf{a}^\top)\mathbf{a} \quad (11)$$

As noted above, when our intervention replaces f with $f_{\mathbf{a}}^*$, we denote the new model as $p_{f \leftarrow f_{\mathbf{a}}^*}$.

We fix f to operate on token-level representations; since $\mathbf{b}$ and $\mathbf{s}$ may have different lengths due to tokenisation; we align representations at the last token of each template region.

In principle, we allow future work to consider other forms of f^* , but 1D DII has two useful properties. Given the linear representation hypothesis and that **CausalGym** exclusively studies binary linguistic features, 1D DII ought to be sufficiently expressive for controlling model behaviour. Furthermore, probes trained on binary classification tasks operate on a one-dimensional subspace of the representation, and thus we can directly use the weight vector of a probe as the parameter $\mathbf{a}$ in eq. (11)—**Tigges et al. (2023)** used a similar setup to causally evaluate probes.

We study seven methods, of which four are supervised: distributed alignment search (DAS), linear probing, difference-in-means, and LDA. The other three are unsupervised: PCA, k -means, and (as a baseline) sampling a random vector. All of these methods provide us a feature direction $\mathbf{a}$ that we use as a constant in eq. (11). For probing and unsupervised methods, we use implementations from scikit-learn (**Pedregosa et al., 2011**). To train distributed alignment search and run 1D DII, we use the pyvene library (**Wu et al., 2024**). Further training details are in appendix B. We formally describe each method below.

4.2 Definitions

DAS. Given a training set T , we learn the intervention direction, potentially distributed across many neurons, that maximises the output probability of the counterfactual label. Formally, we first randomly initialise $\mathbf{a}_{\text{das}}$ and intervene on the model p with it to get $p_{f \leftarrow f_{\mathbf{a}_{\text{das}}}^*}$. We freeze the model weights and optimise $\mathbf{a}_{\text{das}}$ such that we minimise the cross-entropy loss with the target output y_s :

$$\min_{\mathbf{a}_{\text{das}}} \left\{ - \sum_{(\mathbf{b}, \mathbf{s}, y_b, y_s) \in T} \log p_{f \leftarrow f_{\mathbf{a}_{\text{das}}}^*}(y_s \mid \mathbf{b}, \mathbf{s}) \right\} \quad (12)$$

The learned DAS parameters $\mathbf{a}_{\text{das}}$ then define a function $f_{\mathbf{a}_{\text{das}}}^*$ using (11).

Linear probe. Linear probing classifiers have been the dominant feature-finding method for neural representations of language (**Belinkov, 2022**). A probe outputs a distribution over classes given a

representation $\mathbf{x} \in \mathbb{R}^n$:

$$q_{\theta}(y \mid \mathbf{x}) = \text{softmax}(\mathbf{a}_{\text{probe}} \cdot f(\mathbf{x}) + b) \quad (13)$$

We learn the parameters θ of the probe over the base training set examples (so, maximising $q_{\theta}(y_b \mid \mathbf{b})$) using the SAGA solver (**Defazio et al., 2014**) as implemented in `scikit-learn`, and the parameters $\mathbf{a}_{\text{probe}}$ define the intervention function $f_{\mathbf{a}_{\text{probe}}}^*$.

Diff-in-means. The difference in per-class mean activations has been surprisingly effective for controlling representations (**Marks and Tegmark, 2023; Li et al., 2023**) and erasing linear features (**Belrose et al., 2023; Belrose, 2023**). To implement this approach, we take the base input–output pairs $\langle \mathbf{b}, y_b \rangle$ from the training set T , where $y_b \in \{y_1, y_2\}$, and group them by the identity of their labels. Thus, we have $X_1 = \{\mathbf{b} \in T : y_b = y_1\}$ and $X_2 = \{\mathbf{b} \in T : y_b = y_2\}$. The diff-in-means method is then defined as follows:

$$\mathbf{a}_{\text{mean}} = \frac{1}{|X_1|} \sum_{\mathbf{x} \in X_1} f(\mathbf{x}) - \frac{1}{|X_2|} \sum_{\mathbf{x} \in X_2} f(\mathbf{x}) \quad (14)$$

$$= \mu_1 - \mu_2 \quad (15)$$

and as usual $\mathbf{a}_{\text{mean}}$ defines the function $f_{\mathbf{a}_{\text{mean}}}^*$.

Linear discriminant analysis. LDA assumes that each class is distributed according to a Gaussian and all classes share the same covariance matrix Σ . Given the per-class means μ_1 and μ_2 ,

$$\mathbf{a}_{\text{lda}} = \Sigma^{-1}(\mu_1 - \mu_2) \quad (16)$$

Principal component analysis (PCA). We intervene along the first principal component, which is a vector $\mathbf{a}_{\text{pca}}$ that maximises the variance in mean-centered activations (denoted $\tilde{f}(\mathbf{x})$).

$$\max_{\mathbf{a}_{\text{pca}}} \left\{ \sum_{\mathbf{x} \in X_1 \cup X_2} \left(\tilde{f}(\mathbf{x}) \cdot \mathbf{a}_{\text{pca}} \right)^2 \right\} \quad (17)$$

PCA was previously used to debias gendered word embeddings by **Bolukbasi et al. (2016)**.

k -means. We use 2-means and learn a clustering of activations into two sets S_1, S_2 that minimises the variance of the activations relative to their class centroids μ_1, μ_2 . Our feature direction is

$$\mathbf{a}_{\text{kmeans}} = \mu_1 - \mu_2 \quad (18)$$

Mod.	Acc.	Overall odds-ratio ($\uparrow$)								Selectivity ($\uparrow$)							
		DAS	Probe	Mean	PCA	k-m.	LDA	Rand.		DAS	Probe	Mean	PCA	k-m.	LDA	Rand.	
14m	0.62	3.94	1.16	1.04	0.48	0.50	0.11	0.03		1.84	1.38	1.24	0.54	0.55	0.15	0.08	
31m	0.74	5.82	2.22	1.80	0.83	0.85	0.08	0.02		2.75	2.63	2.03	0.86	0.88	0.13	0.03	
70m	0.77	7.60	2.70	2.12	1.16	1.20	0.11	0.03		2.87	2.86	2.15	1.05	1.09	0.16	0.05	
160m	0.82	7.93	3.13	2.23	1.26	1.29	0.12	0.02		2.93	3.27	2.34	1.24	1.26	0.15	0.04	
410m	0.86	10.24	3.69	3.22	2.15	2.19	0.34	0.05		3.96	4.20	3.33	2.07	2.12	0.43	0.06	
1b	0.86	10.74	3.66	3.17	2.07	2.13	0.29	0.03		3.34	4.24	3.09	1.78	1.85	0.36	0.04	
1.4b	0.88	9.58	3.48	3.06	1.96	2.02	0.37	0.02		2.99	4.08	3.21	1.87	1.94	0.46	0.03	
2.8b	0.88	8.88	3.72	3.19	1.93	2.00	0.31	0.01		2.57	4.15	3.31	1.69	1.75	0.39	0.01	
6.9b	0.89	9.95	3.42	2.91	1.81	1.87	0.27	0.01		2.48	3.79	2.85	1.50	1.54	0.34	0.02	

Table 1: Overall odds-ratio (§5.1) and selectivity (§5.2) of each feature-finding method averaged over all tasks in **CausalGym**. We also report average task accuracy, which increases with scale. For models larger than pythia-70m, we report the better of two probes trained with different hyperparameters (appendix C).

5 Experiments

We perform all experiments on the pythia model series (Biderman et al., 2023), which includes 10 models ranging from 14 million to 12 billion parameters, all trained on the same data in the same order. This model series allows us to study how feature representations change with scale and training data size in a controlled manner—all models were trained on the same data in the same order, and checkpoints are provided.

5.1 Measuring causal efficacy

The Transformer (Vaswani et al., 2017) is organised around the **residual stream** (Elhage et al., 2021), which each attention and MLP layer reads from and additively writes to. The residual stream is an information bottleneck; information from the input must be present at some token in every layer’s residual stream in order to reach the next layer and ultimately affect the output.

Therefore, given a feature present in the input and influencing the model output, we should be able to find a causally-efficacious subspace encoding that feature in at least one token position in every layer. If the feature is binary (such as the ones we study in **CausalGym**), then 1D DII should be sufficient for this.

Thus, for each task in **CausalGym**, we take the function of interest f to be the state of the residual stream after the operation of a Transformer layer $l \in L$ at the last token of a particular region $r \in R$. For notational convenience, we denote this function as $f^{(l,r)}$. We learn 1D DII using each method m for every such function. We use a trainset T of 400 examples for each benchmark task, and evaluate on a non-overlapping set E of 100 examples.¹ Each

¹Further training details are given in appendix B, and we report hyperparameter tuning experiments on a dev set in

such experiment results in an intervened model that we denote $p_{f^{(l,r)} \leftarrow f_{\mathbf{a}_m}^*}$. To compute the overall log odds-ratio for a feature-finding method on a particular model on a single task, we take the maximum of the average odds-ratio (§3.4) over regions at a specific layer, and then average over all layers:

$$\text{OverallOdds}(p, m, E) \quad (19)$$

$$= \frac{1}{|L|} \sum_{l \in L} \left(\max_{r \in R} \left(\text{AvgOdds}(p, p_{f^{(l,r)} \leftarrow f_{\mathbf{a}_m}^*}, E) \right) \right)$$

This metric rewards a method for finding a highly causally-efficacious region in every layer.

5.2 Controlling for expressivity

DAS is the the only method with a causal training objective. Other methods do not optimise for, or even have access to, downstream model behaviour. Wu et al. (2023) found that a variant of DAS achieves substantial causal effect even on a randomly-initialised model or with irrelevant next-token labels, both settings where no causal mechanism should exist. How much of the causal effect found by DAS is due to its expressivity? Research on probing has faced a similar concern: to what extent is a probe’s accuracy due to its expressivity rather than any aspect of the representation being studied? Hewitt and Liang (2019) propose comparing to accuracy on a **control task** that requires memorising an input-to-label mapping.

We adapt this notion to **CausalGym**, introducing control tasks where the next-token labels y_b, y_s are mapped to the arbitrary tokens ‘_dog’ and ‘_give’ while preserving the class partitioning.² For example, on the gender-agreement task `agr_gender`, we

appendix C.

²The input-to-label mapping in **CausalGym** tasks is dependent on the input token types, so we cannot exactly replicate Hewitt and Liang. The setup we instead use is from Wu et al. (2023).